

Revisiting Local Area Peer-to-Peer Overlays for Grassroots Social Networks

Final talk for the Master's Thesis by

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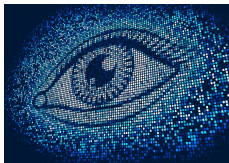
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- Social networks that depend on any party but its members can be shut down.
- Shutdowns, disasters, surveillance and control happen [APP26, Lóp26, SS26, Zub23]
- Use offline-first local communication, augmented with Internet instead of dependent on it



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Introduction

Problems

Π is every agent that could take part. A set $E \subseteq \Pi$ is **essential** if it is the minimal one whose removal leaves every remaining agent unable to interact with anyone; the smallest such $|E|$ sorts every platform into four classes [Sha26, Def. 2.9, 2.10].

Class	Example	Essential agents	cardinality
Centralized	Facebook	the operator's own server [Lóp26]	$ E = 1$
Distributed [JNCL25]			
Federated	Mastodon	instances, and bootstrap nodes:	$ E = \infty, \Pi \setminus E \geq 2$
Blockchain	Steemit	infinite, but not universal [RJC ⁺ 19, LP19]	$1 < E < \infty$
Peer-to-peer	Briar, BitChat	Tor or Nostr relays [osc18, Do25]	$1 < E < \infty$
Grassroots	SSB, GNL	everyone [TLMT19, Sha23]	Π

⇒ **A GSN has $|E| = \Pi \setminus \{p\}$: we chose to model our system after GSNs as universal set, having no central infrastructure**

Introduction

Background: OppNets

- MANETs need a connected end-to-end path, which a crisis zone never offers [CM99, CG14]
- Opportunistic networks **store, carry, forward** until a contact arises [PPC06, CBH⁺07]
- Structured overlays give lower latency and a global state, but lean on designated nodes and degrade under churn [MM02, SMK⁺01, LCP⁺05]
- Unstructured overlays tolerate churn at the price of gossip overhead [LCP⁺05, GTO25]

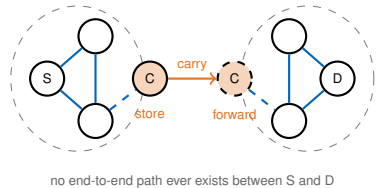


Figure 1: Store, carry, forward

⇒ **Model GSN as OppNet with Store-carry-forward best effort delivery!**

Research question

What must be added to a BLE mesh, without introducing third party infrastructure, for a grassroots social network to deliver at scale?

Contributions

- **Measure** the grassroots networking layer on a testbed of commodity smartphones: what a link costs, how delivery degrades under contention, how often links flap
- **Simulate** at scale, calibrated on those measurements, to compare control-plane and forwarding strategies
- **Results** we found that contention collapses delivery even at close range, link outages are common on old hardware and heavy-tailed, and BLE-only saturates at 80% delivery, with the Internet addition bridging to 100%

System	Local-area underlay	Wide-area underlay	Third-party dependency
SSB [TLMT19]	WLAN, Bluetooth Classic*	pub servers over TCP	pub servers
Briar [osc18]	WLAN, Bluetooth Classic	Tor	Tor [APP26]
BitChat [Do25]	BLE mesh	Nostr relays	the relays [WT25]
Amigo [IRJ+25]	Wi-Fi Direct	none	GPS

- The component structure inherits from BitChat; Nostr relays replaced by edge rendezvous points (friends) for hole-punching

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Background

The local-area underlay

Six criteria: interoperability and availability on commodity-grade smartphones, low power, works while backgrounded, tolerates mobility, needs no infrastructure, and carries a short message within a second [Nie94]

Technology		Why
Wi-Fi Ad-Hoc	✗	no power saving when idle [FKK13]
Wi-Fi Direct, AirDrop, QuickShare	✗	requires a foreground app; no interoperability between iOS and Android [CMGSS13]
Wi-Fi Aware	✗	reached iOS only in 2025 [App25]
Zigbee, LoRa, 6LoWPAN	✗	no device-to-device support on smartphones [SH16]
Bluetooth Low Energy	✓	on all smartphones; duty-cycled radio; the OS advertises and scans in the background; about 200 kb/s [TTS ⁺ 17]

The transport must survive a change of network attachment point, and cost as little as possible to (re-)establish, no encryption or handshake

Transport		Why
TCP	✗	the connection breaks when its 5-tuple changes, congestion control reads an outage as congestion [Edd22, APB09], and throughput falls with the hop count [MIF02]
Plain UDP	✗	survives address changes, but offers no reliability, ordering, or congestion control [Pos80]
QUIC	○	reliable and congestion-controlled, but pays a handshake and brings encryption the overlay already has [LRW ⁺ 17]
UDX	✓	Similar to QUIC, but no handshake and no encryption [Hol26]

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System Design

One overlay, two underlays

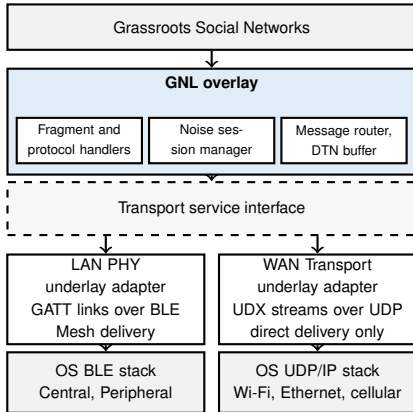


Figure 2: GNL component architecture

- GNL fragments to the path MTU, seals every fragment in the peer's Noise session, and picks the underlay that currently reaches the peer
- Radio first: a peer reachable over both is served over BLE
- No link to the recipient? The router takes custody and conveys the packet over the mesh

8 B fixed prefix SHA-256("grassroots")	8 B rotating suffix , new every 15 minutes SHA-256("grassroots ble suffix" pubkey slot)
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Figure 3: The advertised BLE service UUID

- **Identity** is an Ed25519 key pair: the address, the signature, and the root of the session key
- **ANNOUNCE** broadcasts presence and identity
- Friends who know the key recognize the beacon; random to strangers and cannot be tracked across slots
- **Trust**: **Closed** links only to friends and never relays; **Open** links to any grassroots device and joins the mesh

System Design

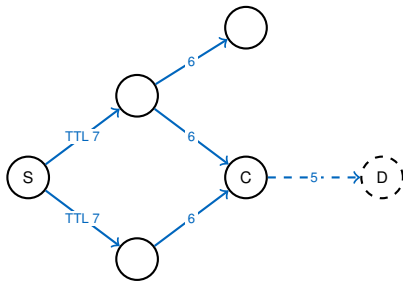
Secure sessions and the packet header

- **Noise XX** with Curve25519 and ChaCha20-Poly1305 [KNB18, Vla16]: a static key that differs from the announced one aborts the handshake
- One session per peer across both underlays, keyed by public key
- 60-byte outer envelope header with a recipient but **no sender**: relays read only the envelope
- The receiver finds the sender by trial decryption against its sessions, to read inner encrypted envelope

System Design

Routing over the BLE mesh

- **Controlled flooding:** forward packets using GCS buffer reconciliation
- **Custody:** keep packets per recipient for 6 hours or until a delivery proof arrives
- **Fragmentation:** the 247-byte MTU leaves 138 B of payload per fragment [TPHB21]



Duplicates are suppressed by packet ID. C keeps custody until it links with D or sees a delivery proof.

Figure 4: Controlled flooding with custody

System Design

The IP path: rendezvous through friends

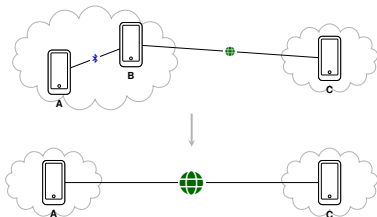


Figure 5: B, a mutual friend linked to A over BLE and to C over the Internet, signals the hole punch; afterwards A and C communicate directly

- NATs drop unsolicited packets, so friends need a **signaling channel** to punch holes [SFK08]: a direct BLE link, or mutual online friends as rendezvous agents
- The rendezvous agent reflects each peer's public address and instructs both to send punch packets; signed **friendship attestations** authorize it
- A UDX stream opens, then Noise XX authenticates it
- **Cold calls:** a signed, single-use invite link, valid for limited time

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- Eight commodity smartphones on tripods, outdoors
- Every run traces RSSI, the time to each link stage, and bytes per packet class
- Messages are 138 B, the largest unfragmented payload
- Three experiments: the [line](#) (distance), [diluting clique](#) (contention), [link flapping](#) (outages)

Evaluation

Experiment 1, the line: the path-loss fit

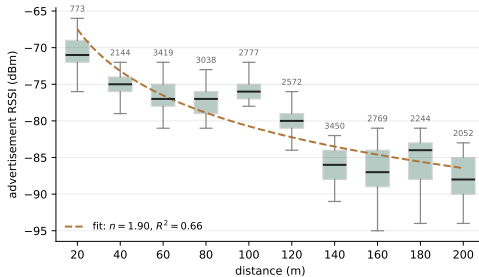


Figure 6: Path loss vs. distance

⇒ $\text{RSSI}(d) = -43.5 - 18.8 \log_{10} d$, **exponent** $n \approx 1.9$, **residual** $\sigma = 3.6$ dB

- Compare to Rappaport standard path-loss model for 5G waves [MZNR13]
- Two Nexus 5X from 20 to 200 m in 20 m steps; ten cold-start runs per point, 100 messages each way

Evaluation

Experiment 1, the line: time to each stage



Figure 7: The five link stages, with median time from a cold start

- Discovery is bound by the radio; every later stage is paid for by the application

⇒ **Identifying the peer is the costliest app step at 0.84 s, the Noise handshake adds only 0.2 s.**

Evaluation

Experiment 1, the line: bytes to each stage

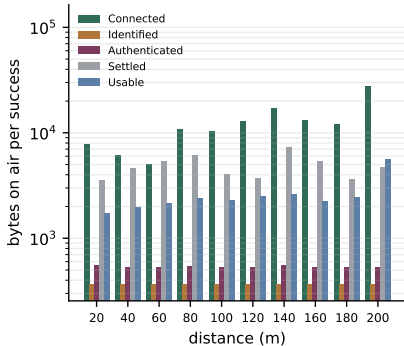


Figure 8: Bytes to each stage, by distance

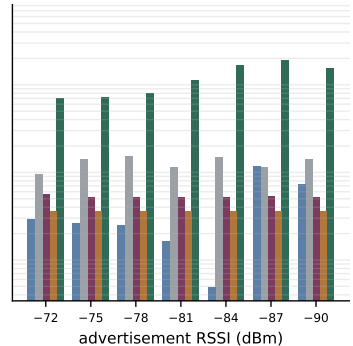


Figure 9: The same runs, by RSSI

- Advertising dominates the other stages: reaching a GATT link costs up to 27 kB at a 135 ms advertisement interval, followed by session (524 B) and identity (364 B)

⇒ Finding the peer is what costs bytes; the secure session itself is nearly free.

Evaluation

Experiment 2, the diluting clique: contention collapses delivery

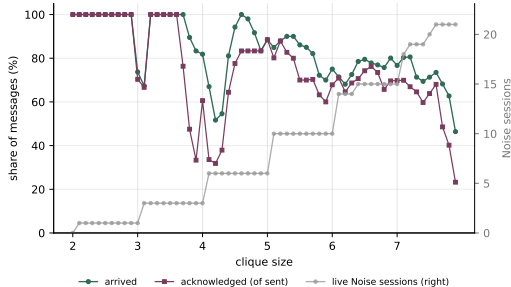


Figure 10: Delivery and ACKs per trial as the clique grows

- The clique grows from 2 to 7 phones within 30 m; offered load rises from 6.7 to 140 msg/s
- Delivery falls from 100 % to 50 %, ACKs to 20 %, median latency from 110 ms to 9 s
- Each join degrades delivery, as all nodes spend airtime on link establishment

⇒ **Radio contention and hardware, not the range, are the constraints: every pair sat at a lossless RSSI.**

Evaluation

Experiment 3, link flapping: frequent link loss, heavy-tailed outages

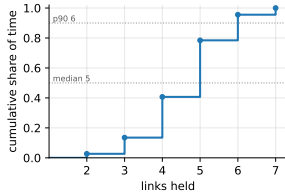


Figure 11: Links held, CDF

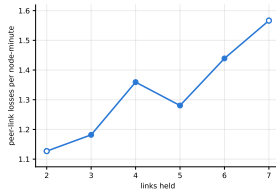


Figure 12: Link loss rate vs. links held

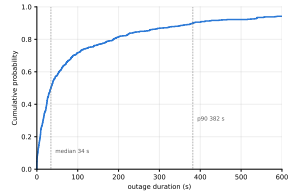


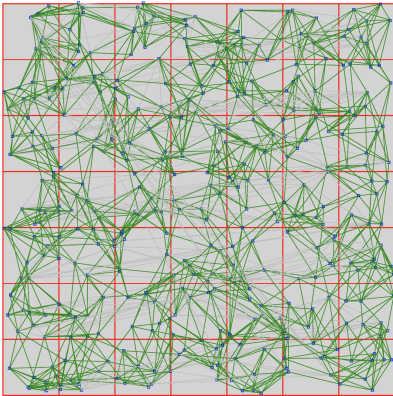
Figure 13: Outage duration, CDF

- Eight phones at short range with sessions established beforehand: only physical link loss is measured

⇒ **These distributions calibrate the simulation.**

Evaluation

Simulations: congestion at scale



Red: the 49 cells of 50 m, 8 nodes each. Green: the lanes messages flood on; grey: links held but passive

Scenario in the ONE [KOK09]

- 350×350 m hall, 49 cells, 8 nodes per cell: 392 nodes
- 50 m radio range (lossless), at most 7 links per node
- One new message per second for 30 000 s; 25 seeded placements per arm

Five control-plane establishment strategies

flood	epidemic flooding, as in GNL and BitChat
f3	flood only the top 3 links by ego-network centrality [DH07] and buffer occupancy
oracle	omniscient shortest path: the BLE ceiling
amigo	leader-elected territories, after Amigo's dynamic clique routing [IRJ ⁺ 25]
amigo-udx	amigo plus UDX between leaders, at the 2.4 MB/s measured on a Nexus 5X

Evaluation

Simulations: the model the arms run on

Link flapping, calibrated on experiment 3. Every $\Delta t = 30$ s a node u holding k_u links drops $D_u \sim \text{Pois}(\lambda(k_u) \Delta t)$ of them, chosen uniformly. Endpoints draw independently, so link i – j flaps at

$$\lambda_{ij} = \frac{\lambda(k_i)}{k_i} + \frac{\lambda(k_j)}{k_j} \quad k_i = k_j = 7 : \quad \lambda_{ij} \approx 0.45 \text{ min}^{-1}, \text{ mean time between flaps } 2.2 \text{ min}$$

and stays down at both ends for a resampled measured outage: heavy-tailed, median ≈ 35 s.

Simulations: only the wide-area underlay overcomes the hop limit

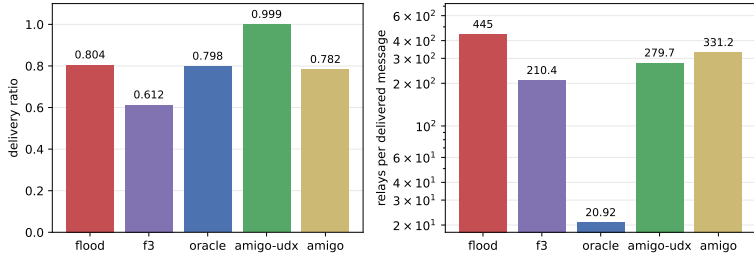


Figure 14: Delivery ratio and relays per delivered message, five arms

- flood and oracle both deliver about 80 %: the hop limit of 7 bounds reachability. Flooding costs 445 relays per delivered message, the oracle 21
- f3 halves the relays but falls to 61 %; amigo on BLE alone reaches 78 % at near-flood cost
- amigo-udx delivers 99.9% with a third fewer relays than flooding and the lowest buffer occupancy

⇒ **A BLE-only mesh does not scale to this network, not even with optimal routing. Rendezvous points should be used as backbone to deliver messages across the network.**

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Can BLE alone carry a grassroots social network?

Not at scale: delivery is bounded by the radio range under a hop limit, and by the airtime a node shares among its links.

⇒ **In a full shutdown, form links deliberately, guided by RSSI. Whenever an Internet path survives, bridge the mesh across it.**

Conclusion

Future work

- **Control plane:** a user-chosen scenario policy, emergency or protest, that sets link admission by RSSI and the flooding budget
- **Buffers:** VACCINE-style antipackets [LPO10, LPO11] that purge delivered messages, weighed against the presence they reveal
- **Handshakes:** Noise patterns for friends who already know each other's keys save a message and keep keys off the air
- **NAT traversal and BLE stack:** a grassroots AutoNAT through several friends; connection intervals, scan windows, stale-address failures
- **Beyond messaging:** opportunistic networks as a substrate for collective communication on co-located devices

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- At close range, **Contention binds the BLE mesh**. Delivery falls with the number of active links per node, at an RSSI where a single link is lossless
- **A second transport between rotating leaders is what scales**, from 80 % to 99.9 %. Leaders can double as gateways and rendezvous agents, and rotate as they drain
- **The price is exposure**. Leaders and rendezvous points attract traffic analysis and denial of service; onion-style hops would trade usability for privacy [APP26]
- **Judge links by RSSI, not by geography**, unlike Amigo's GPS grid and CityMesh's building map [IRJ+25, LCL+26]
- **Limits**. Leader election is free in the simulation; greedy linking is the wrong default in dense settings

⇒ **Thank you! questions?**

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Backup

Experiment 1, the line: RSSI, not distance, predicts the cost

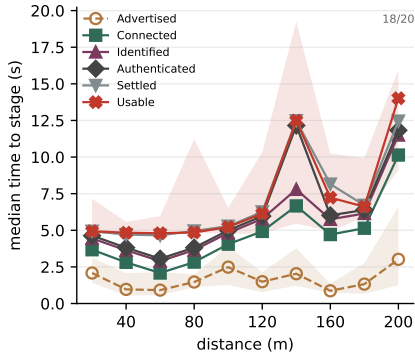


Figure 15: Time to link stage, by distance

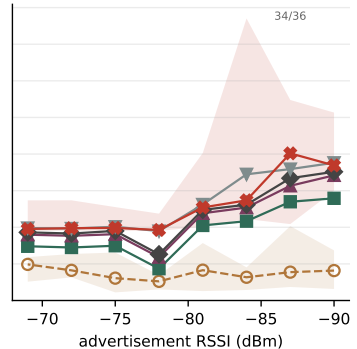


Figure 16: The same runs, by RSSI

- Discovery is quick and bound by the radio; identifying the peer is what costs.
- By distance the picture is confusing: RSSI values measured at different distances overlap

⇒ **Pooled by RSSI, every stage rises monotonically.**

Backup

Experiment 1, the line: delivery and acknowledgements

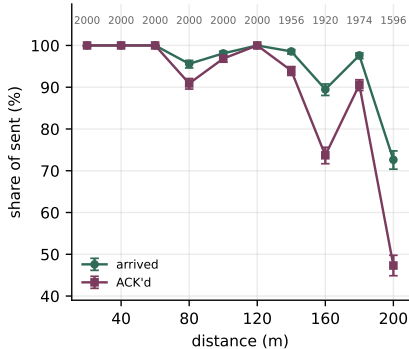


Figure 17: Delivery and ACKs, by distance

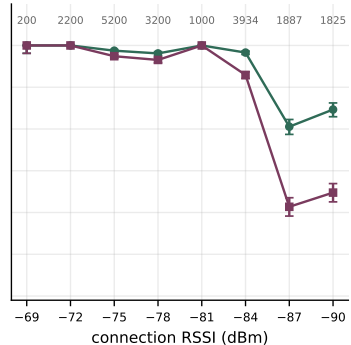


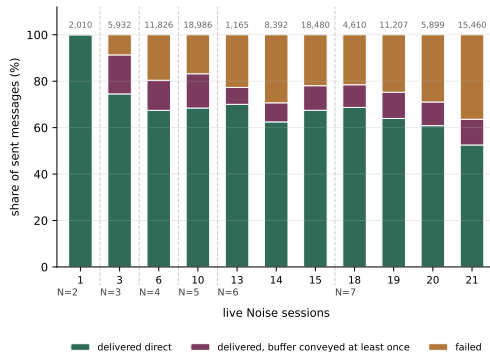
Figure 18: The same runs, by RSSI

- Lossless above -84 dBm

⇒ Judge links by RSSI.

Backup

Experiment 2, the diluting clique: custody does not rescue delivery



- Per message: delivered directly, conveyed from a peer's buffer, or failed
- Direct delivery drops to about 50 % at 21 live sessions; failures rise from 9 % to 37 %
- Conveyance covers only 5 to 10 % of deliveries

Figure 19: Per-message outcome by live Noise sessions

⇒ **Greedy link establishment fails under contention: control plane link establishment needs a strategy.**

Backup

Experiment 3 the link-flap model in numbers

Nexus 5X link losses used by the simulator

links held k	$\lambda(k)$ per node-minute	per link, $\lambda(k)/k$
1	1.13	1.13
2	1.13	0.57
3	1.18	0.39
4	1.36	0.34
5	1.28	0.26
6	1.44	0.24
7	1.57	0.22

$k = 1$ was never observed and reuses the value of $k = 2$

- The per-node rate rises mildly with k , while the per-link rate falls: a node with more links is not proportionally more fragile per link
- Every 30 s each node draws a Poisson number of drops and picks the victim links uniformly
- Outage lengths are resampled from 1 000 quantiles of the measured Nexus 5X outages: median 37 s, mean 236 s, maximum 5 356 s
- An exponential fit was rejected: too few short and too few very long outages

Backup

Delivery under load along the line

d (m)	sent	arrived	acked	arrived	acked	ack p50
20	2000	2000	2000	100.0 %	100.0 %	123 ms
40	2000	2000	2000	100.0 %	100.0 %	133 ms
60	2000	2000	2000	100.0 %	100.0 %	135 ms
80	2000	1912	1818	95.6 %	95.1 %	150 ms
100	2000	1962	1937	98.1 %	98.7 %	144 ms
120	2000	2000	2000	100.0 %	100.0 %	166 ms
140	1956	1928	1838	98.6 %	95.3 %	380 ms
160	1920	1719	1415	89.5 %	82.3 %	1919 ms
180	1974	1929	1788	97.7 %	92.7 %	435 ms
200	1596	1161	755	72.7 %	65.0 %	3138 ms
all	19446	18611	17551	95.7 %	94.3 %	154 ms

100 messages per trial per direction, both directions pooled; ack latency is the per-trial median of medians

Delivery and latency along the line, by distance

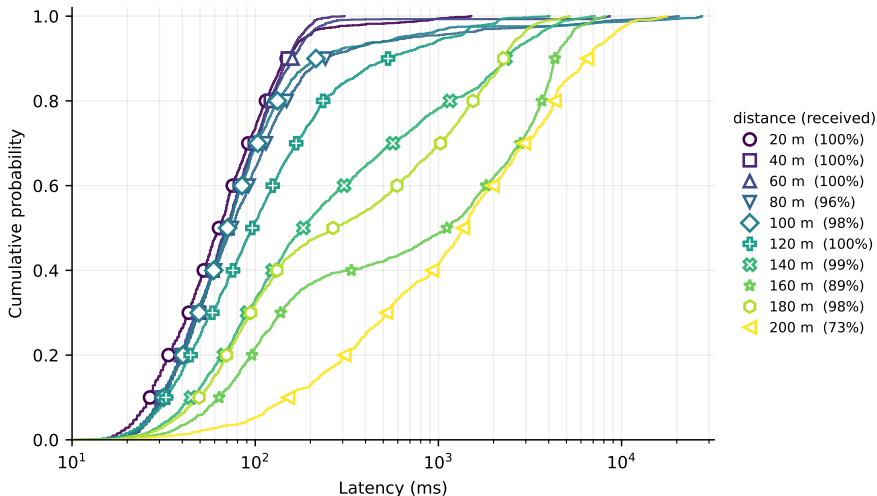


Figure 20: Message latency CDF, by distance

N	offered	sent	arrived	acked	lat p50	carried
2	6.7/s	2000	100.0 %	100.0 %	0.11 s	0.0 %
3	20.0/s	5965	91.5 %	79.5 %	11.0 s	5.4 %
4	40.0/s	11940	80.3 %	63.9 %	31.6 s	10.1 %
5	66.7/s	19736	83.6 %	75.1 %	12.3 s	7.6 %
6	100.0/s	28793	75.7 %	70.3 %	8.3 s	6.5 %
7	140.0/s	38112	70.3 %	57.8 %	9.1 s	10.4 %

Ten trials per clique size, pooled over all senders; offered load in messages per second. **Carried**: share of deliveries made out of a store-carry-forward buffer rather than over a live link.

Advertising cadence at idle

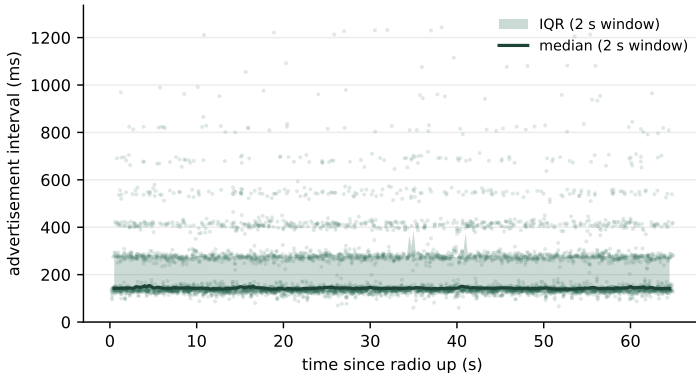


Figure 21: Median 7.41 Hz, an advertisement interval of about 135 ms, over twenty cold desk runs; even with an idle scanner roughly a third of the advertisements are not captured. Native stacks expose no callback for sent advertisements, so this baseline estimates the advertising bytes on the line.

Backup

Raw capacity of the two underlays on a Nexus 5X

BLE, one GATT link, 1 m, write without response

Write size (B)	Writes/s	Delivered (kB/s)
64	202	12.9
138	222	30.7
244	223	54.5
380	165	62.7
448	145	65.0
514	125	64.1

The binding constraint is writes per second, not bytes offered;
the ceiling is 63 to 65 kB/s at the full 514-byte ATT payload

UDX, one isolate per stream

Concurrent peers	Total throughput (bytes/s)
1	628 800
2	1 321 300
4	1 952 600
6	1 634 900
20	2 459 200
48	2 406 300
96	1 757 500

About 2.4 MB/s, the UDX capacity used by the amigo-udx arm

Buffer occupancy at scale

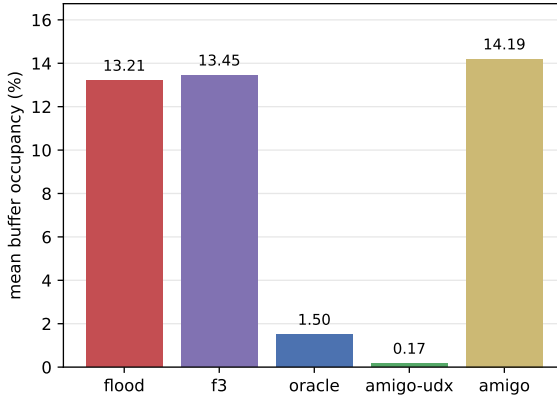


Figure 22: Mean DTN buffer occupancy for the five arms at 392 nodes, 1 msg/s network-wide, 25 seeded placements per arm. amigo-udx needs the least buffer; f3 relays half as much as flood but buffers the same; amigo over BLE alone buffers the most.

Backup

Noise XX in three messages

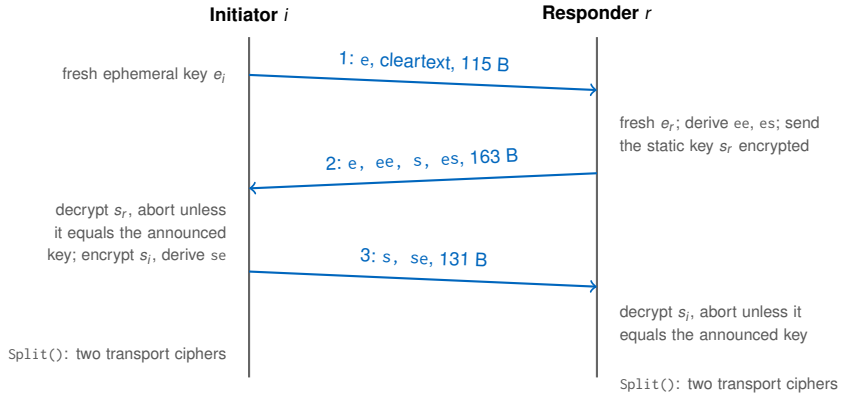
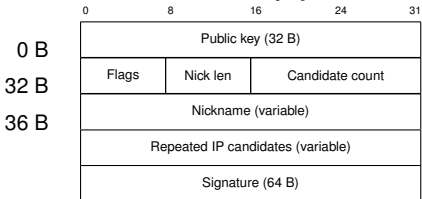


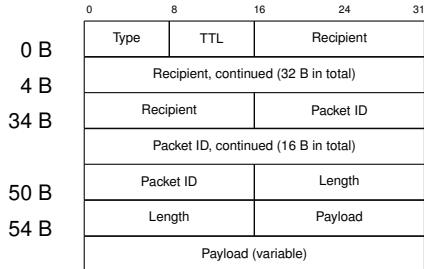
Figure 23: e : send a fresh ephemeral key. s : send the static key. ee , es , se : Diffie-Hellman between the named keys, mixed into the chaining key with HKDF. Both peers send message 1; the lexicographically lower public key takes the responder role.

ANNOUNCE payload



Sent in cleartext with TTL 1; signed by the creator

GrassrootsPacket header, 60 B



No sender, no signature: the Noise session authenticates the sender; the recipient field avoids trial decryption of transit packets

Crypto testbench: the cost of trial decryption

active sessions	per transit packet	packets/s to saturate one core	CPU at 10 packets/s
8	2.93 ms	341	2.9 %
32	11.5 ms	87	11.5 %
128	44.8 ms	22	44.8 %

- Measured on a Nexus 5X: trial decryption of one transit packet against a growing number of active Noise sessions
- The header carries no sender, so the receiver finds it by trial decryption, most recently used session first
- Without a recipient field every transit packet would pay this cost: an active user with 128 sessions saturates one core at 22 packets per second, and spends almost half a core at only 10
- Hence the header keeps the recipient and omits the sender

Units	Join order	Device	SoC	OS	Bluetooth
4	1–4	LG Nexus 5X	Snapdragon 808	Android 8.1	4.2
2	5–6	Google Pixel 10 Pro	Tensor G5	Android 16	6.0
1	7	Samsung Galaxy S10e	Exynos 9820	Android 12	5.0
1	8	Huawei HMA-L09	Kirin 960	Android 9	4.2

- All phones stand on tripods outdoors and interoperate at Bluetooth 4.2
- The join order is the order in which devices enter the diluting clique
- The Huawei is the eighth device, used in the link flapping experiment